Xylem Pipeline Relay

Move water uphill the way a tree does

At a glance: Move water uphill like a tree, faster and cleaner than rival teams. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: capillary action, plant water transport, material choice, geometry (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- paper towels: about 6 sheets per team per run (about 1 roll for the camp)
- cotton string: cut into about 12 in wicks, 3 per team per run (1 ball)
- felt strips: about 1 × 12 in, 3 per team plus spares (16 total; source locally, not on the buy line)
- cups: 2 per team (source and target), 8 plus 4 spare
- food coloring: washable, a few drops per cup (1 to 2 small bottles; source locally, not on the buy line)
- trays: 1 per team (4, plus 1 spare)
- rulers: 1 per team (4, on hand)
- elevation blocks: 1 per team (4, on-hand blocks or books)

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

01 Set up one station per team with the materials above, sorted and ready.

02 Mark each target cup with a fill/read line (or stage a kitchen scale), mix the colored source water, stage a fill jug and pour-off bucket, and put out the scoring sheet before testing begins.

03 Put out a containment tray at each station and have gloves available for students who do not want dye-stained hands. No goggles are needed: this is moving water with washable dye, not a chemical or splash hazard.

04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

`0:00 - 0:05`

Pose the mission as a challenge: "Move water uphill like a tree, faster and cleaner than rival teams." Take a quick prediction by show of hands.

Phase 2 Prediction

`0:05 - 0:12`

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

`0:12 - 0:18`

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve

`0:18 - 0:44`

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Test and record

Teams run the baseline transfer for the set time limit over the tray, read water delivered against the marked cup or scale, and log it before changing anything.

Phase 6 Redesign

Each team changes exactly one variable (path length, slope, fiber contact, or wick material), predicts the effect, and re-runs for the same time limit to compare against baseline.

Phase 7 Score, debrief, cleanup

`last 12 min`

Teams submit a score slip, answer a debrief question with evidence, then reset the station: pour dyed water into the pour-off bucket, bin used wicks, rinse and wipe cups and trays, dry the table, and wash hands (remove gloves first if worn).

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: Little or no water reaches the target on the first run. Fix: Check that the wick is in firm contact along the whole route and both ends are seated in the water, that the slope is not too steep, and that the source cup is filled to the line; log the baseline anyway.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to capillary action.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which material moved water best, and why?
- How did slope change your delivery?
- Where does a real tree use this same effect?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Water delivered	40
Speed	20
Least spill	15
Design explanation	15
Redesign gain	10
Total	**100**

CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

SOURCES Science Buddies; plant water transport references